

Quantum Toom's Contours

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Introduction

Today:

- ▶ Proof.
 1. Walkthrough the math

Our Code

Code

Consider the representation of the d -dimensional toric by the Cayley graph of the $(\mathbb{Z}_n^d, +)$ group generated by the elements $S = \{1_i : i \in [d]\}$ ¹ We say that an edge $\{x, y\}$ is positive relative to a plaquette $\{v, g_1 v, g_2 g_1 v, g_2 v\}$, defined by a rooted vertex $v \in \mathbb{Z}_n^d$ and the generators $g_1, g_2 \in S$, if $x = v$ and y is either $g_1 v$ or $g_2 v$. In our code, bits are set on the plaquette, and the edges correspond to checks that restrict the parity of the bits on their support to be even.

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1 \Delta_2$.

¹where 1_i is the vector consisting of one at the i th coordinate and zero elsewhere

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$ -depolarized channel:

- ▶ There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- ▶ There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift-rule and the flip upon majority are local decoders.